



PES UNIVERSITY

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Department of Computer Science and Engineering

UE24MA241B – Linear Algebra and Its Applications

Sample Mini Project

Movie Recommendation System Using a Linear Algebra Pipeline

Problem Statement

Online streaming platforms recommend movies by analyzing user preferences. In this mini project, you will build a simplified movie recommendation system using a structured linear algebra pipeline starting from raw rating data and ending with predicted ratings and trend discovery. The goal is to predict how a user might rate movies they have not watched yet.

Linear Algebra Flow Applied to Movie Recommendation

Step 1: Real-World Data to Matrix Representation

We collect ratings given by users to movies. Rows represent users, columns represent movies, and each entry represents a rating. This forms the User–Movie Rating Matrix A .

Topics Used:

Matrices

Systems of Linear Equations

Matrices as Linear Transformations

Step 2: Matrix Simplification

We analyze the structure of ratings by reducing the matrix. The matrix is converted into Row Reduced Echelon Form (RREF) to identify independent rating patterns.

Topics Used:

Gaussian Elimination

Row Reduced Echelon Form

LU Decomposition (optional computational method)

Step 3: Structure of the Space

We study the spaces formed by the rating matrix. The row space represents the user preference space. The column space represents the movie feature space. The null space represents hidden or missing preference patterns.

Topics Used:

Vector Spaces and Subspaces

Basis and Dimension

Rank and Nullity

Column Space and Null Space

Step 4: Remove Redundancy

Some users may have similar tastes, leading to dependent vectors. We extract a basis that represents independent taste directions.

Topics Used:

Linear Independence

Basis of a Vector Space

Step 5: Orthogonalization

We convert the basis into orthogonal preference directions so each represents a distinct taste factor such as action, romance, or comedy.

Topics Used:

Orthogonal Vectors

Orthogonal Bases

Gram–Schmidt Orthogonalization

Step 6: Projection

To predict a missing rating for a user, we project their known ratings onto the movie feature subspace. This gives the closest consistent rating estimate.

Topics Used:

Orthogonal Projections

Projection onto Subspaces

Step 7: Prediction Using Least Squares

Because the system may be inconsistent due to missing entries, we compute the best approximate solution using least squares.

$$\hat{x} = (A^T A)^{-1} A^T b$$

Topics Used:

Least Squares Solution

Matrix Transpose and Inverse

Step 8: Pattern Discovery Using Eigenvalues

We compute eigenvalues of the covariance matrix of ratings to identify dominant viewing trends. Large eigenvalues correspond to the strongest common taste patterns among users.

Topics Used:

Eigenvalues and Eigenvectors

Step 9: System Simplification

Using eigenvectors, we reduce the system to only the most important preference directions. This improves efficiency and removes noise from the recommendation model.

Topics Used:

Diagonalization of a Matrix

Diagonalization of Symmetric Matrices